



TARGET-90

JKBOSE CLASS-11TH

SEQUENCES AND SERIES

One Shot

MATHS

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Topics To Be Covered

- 1 ... Sequence and Series.
- 2 ... Geometric Progression
- 3 ... PYQ + Important Questions.



SEQUENCES AND SERIES



What is a Sequence?

A sequence is an **ordered list of numbers**, arranged according to a particular rule.

Example:

2, 5, 8, 11, 14, ...

(Each term increases by 3)

What is a Series?

A series is the **sum** of the terms of a sequence.

Example:

2 + 5 + 8 + 11 + 14 + ...

(Sum of the terms of the sequence)

Types of Sequences

- 1 Arithmetic Sequence**
(Constant difference)
2, 5, 8, 11, 14, ... ($d = 3$)
- 2 Geometric Sequence**
(Constant ratio)
3, 9, 27, 81, ... ($r = 3$)
- 3 Harmonic Sequence**
(Reciprocal of an arithmetic sequence)
 $1, \frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \frac{1}{5}, \dots$

Arithmetic Progression (AP)

n^{th} term: $a_n = a + (n - 1)d$

Sum of first n terms:

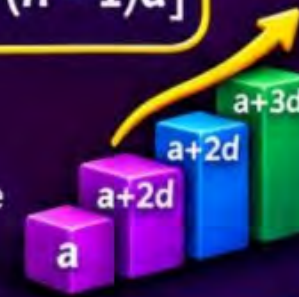
$$S_n = \frac{n}{2} [2a + (n - 1)d]$$

Where,

a = first term

d = common difference

n = number of terms



Geometric Progression (GP)

n^{th} term: $a_n = ar^{n-1}$

Sum of first n terms ($r \neq 1$):

$$S_n = \frac{a(r^n - 1)}{r - 1}$$

Where,

a = first term

r = common ratio

n = number of terms



Sum of First n Natural Numbers

$$1 + 2 + 3 + \dots + n = \frac{n(n+1)}{2}$$

Sum of Squares:

$$1^2 + 2^2 + 3^2 + \dots + n^2 = \frac{n(n+1)(2n+1)}{6}$$

Sum of Cubes:

$$1^3 + 2^3 + 3^3 + \dots + n^3 = \left[\frac{n(n+1)}{2} \right]^2$$

Important Formulas

- a_n (AP) = $a + (n - 1)d$
- S_n (AP) = $\frac{n}{2} [2a + (n - 1)d]$
- a_n (GP) = ar^{n-1}
- S_n (GP) = $\frac{a(r^n - 1)}{r - 1}$ ($r \neq 1$)
- $S_n = \frac{n(n+1)}{2}$
- $S_n^2 = \frac{n(n+1)(2n+1)}{6}$
- $S_n^3 = \left[\frac{n(n+1)}{2} \right]^2$

Key Points

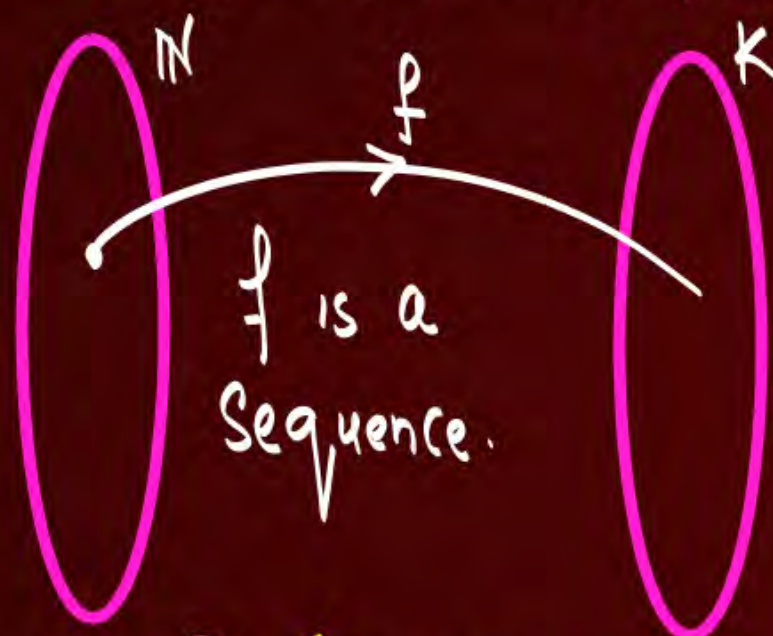
- ✓ A series is the **sum** of terms of a sequence.
- ✓ AP has constant **difference**.
- ✓ GP has constant **ratio**.
- ✓ The n^{th} term and sum formulas help us find any term or sum without writing all the terms.
- ✓ These concepts are widely used in real life (finance, science, growth, etc.).



Practice + Formulas = Success

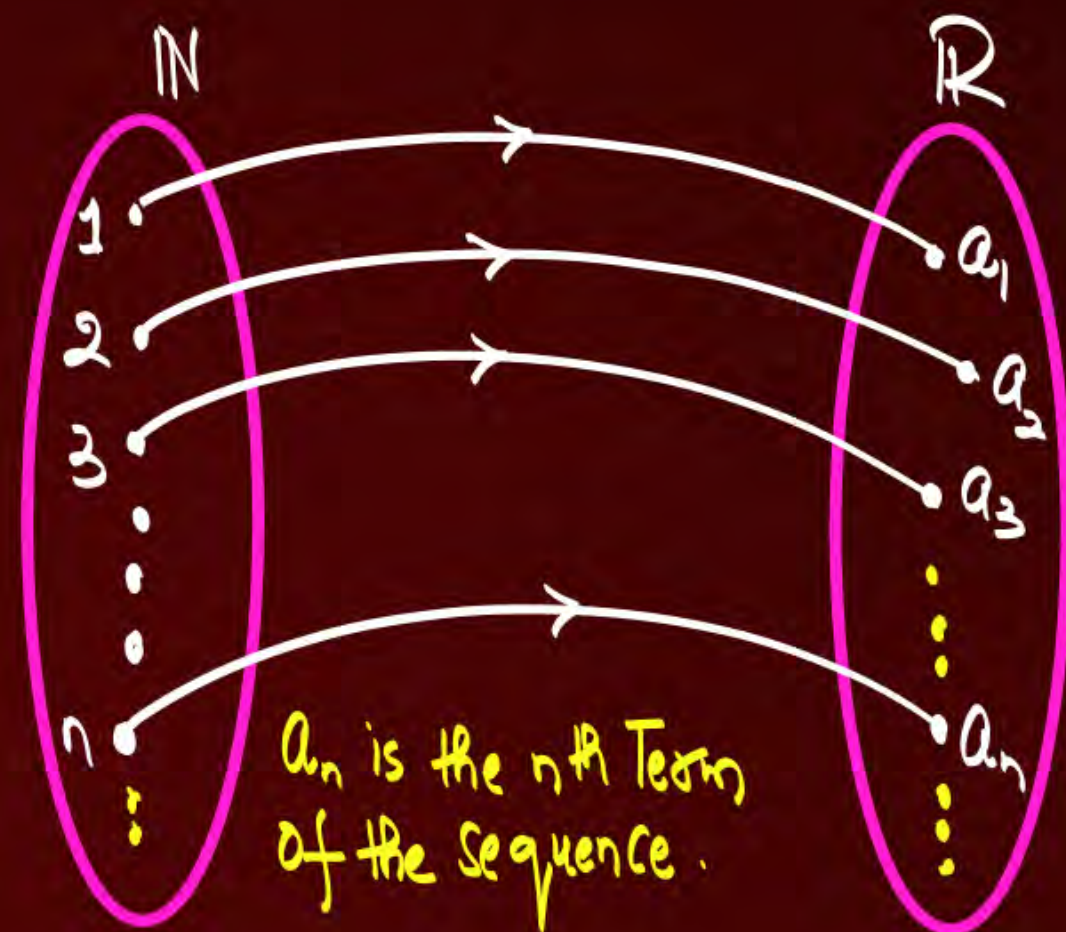
INTRODUCTION

Definition: A function whose domain is the set of Natural numbers is known as a Sequence (Infinite Sequence)



* $f: \mathbb{N} \rightarrow \mathbb{R}$ (Real Sequence)

* $f: \mathbb{N} \rightarrow \mathbb{C}$ (Complex Sequence)



* $1, -1, \frac{1}{2}, -\frac{1}{2}, \frac{1}{9}, \frac{1}{11}, \dots$

$\downarrow \quad \downarrow \quad \downarrow \quad \downarrow \quad \downarrow \quad \downarrow$

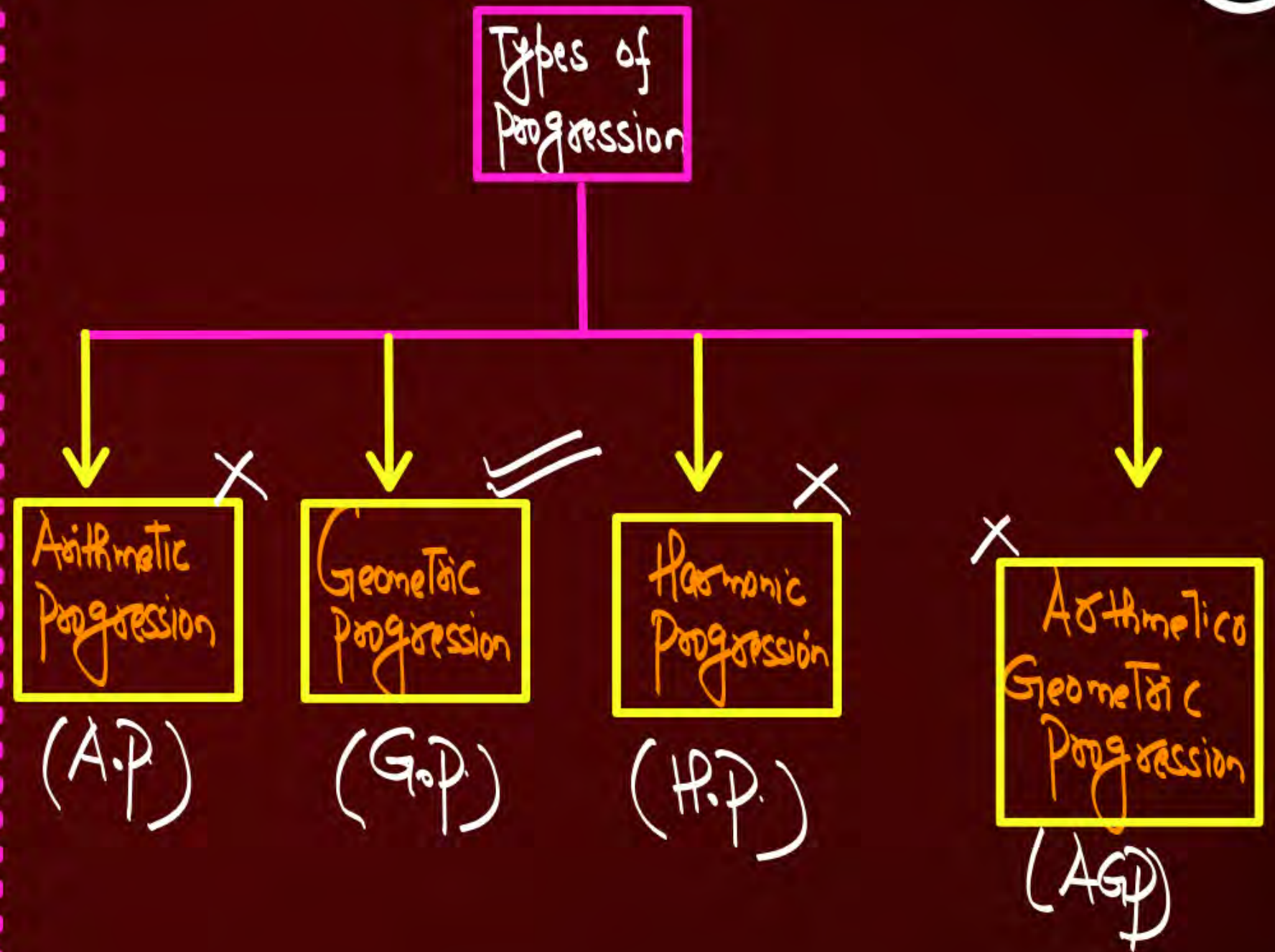
$a_1 \quad a_2 \quad a_3 \quad a_4 \quad a_5 \quad a_6 \dots$

★ **Progression**: A Sequence with a Pattern is known as a Progression.

* 2, 4, 6, 8, 10, 12, ... (Progression)
 $a_n = 2n \quad n \in \mathbb{N}$. (Arithmetic Progression)

* 1, $\frac{1}{2}$, $\frac{1}{3}$, $\frac{1}{4}$, ... (Progression)
 $a_n = \frac{1}{n}$

A Sequence whose all Terms Can be obtained by a General formula is known as a Progression.



EXAMPLE



Write the first three terms in each of the following sequences defined by the following:

(i) $a_n = 2n + 5,$

(ii) $a_n = \frac{n-3}{4}, a_1 = -\frac{2}{4} = -\frac{1}{2}.$

$$a_n = 2n + 5$$

$$a_1 = 2 \times 1 + 5 = 7$$

$$a_2 = 2 \times 2 + 5 = 9$$

$$a_3 = 2 \times 3 + 5 = 11.$$

EXAMPLE



A **sequence** may be **defined** as a

- (a) Relation, whose range $\subseteq \mathbb{N}$ (natural number) ✗
- (b) Function whose range $\subseteq \mathbb{N}$ ✗
- (c) Function whose domain $\subseteq \mathbb{N}$ ✓
- (d) None of the above ✗

QUESTION

The 18th term of the sequence defined by $a_n = \frac{n(n-3)}{n+4}$

1 $\frac{140}{11}$ ✗

2 $\frac{135}{11}$ ✓

3 $\frac{135}{12}$ ✗

4 $\frac{140}{16}$ ✗

$$a_n = \frac{n(n-3)}{n+4}$$

$$\Rightarrow a_{18} = \frac{18 \times 15}{22} = \frac{135}{11}$$

Geometric Progression (G.P.)

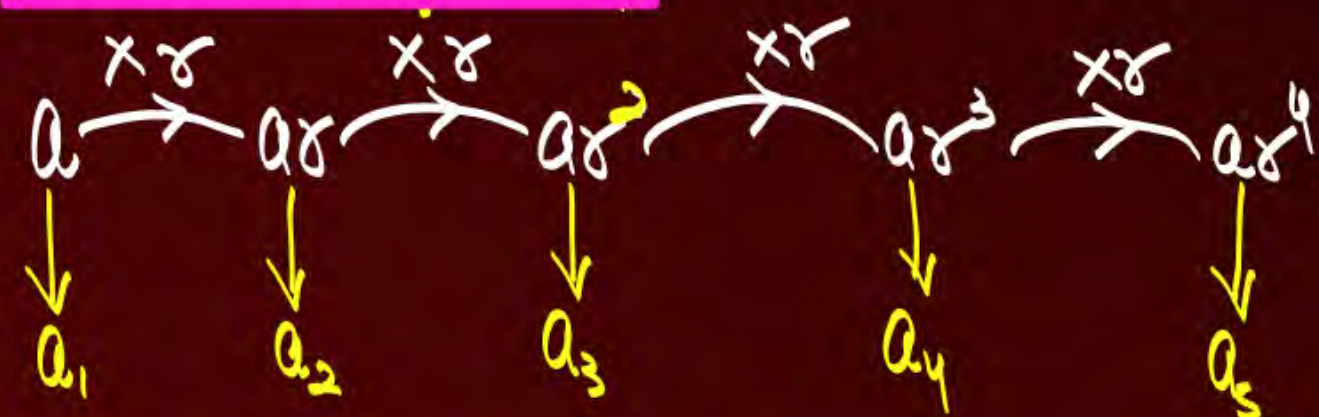
★ **Definition** A Sequence of Non-zero Terms whose each Succeeding Term can be obtained from the preceding Term by multiplying a Constant, is known as Geometric Progression.

2 $\xrightarrow{\times 2}$ 4 $\xrightarrow{\times 2}$ 8 $\xrightarrow{\times 2}$ 16 $\xrightarrow{\times 2}$ 32, (Geometric Progression)

* The Ratio of the Successive Terms is Constant

* This Constant is usually denoted by r and is known as the Common Ratio.

★ **General Term of a G.P.**



Based on the above observation, we can see that

$$a_n = ar^{n-1}$$

* let S_n denote the Sum of n -terms of a G.P.

$$\text{then } S_n = \begin{cases} na ; r=1. \\ \frac{a(r^n-1)}{r-1} ; r \neq 1. \end{cases}$$

EXAMPLE

Find the 10^{th} and n^{th} terms of the G.P. 5, 25, 125,

(Sol) Given G.P. is: 5, 25, 125, ...

Here $a = 5$ and $r = 5$. So

$$\text{that } a_n = ar^{n-1} = 5 \times 5^{n-1}$$

$$\Rightarrow a_n = 5^n$$

$$\& a_{10} = 5^{10}$$

EXAMPLE

(Important)

How many terms of the G.P. $3, \frac{3}{2}, \frac{3}{4}, \dots$ are needed to give the sum $\frac{3069}{512}$?

(Sol) Given G.P: $3, \frac{3}{2}, \frac{3}{4}, \frac{3}{8}, \dots$

Here $a = 3$, $r = \frac{1}{2}$. Suppose that $S_n = \frac{3069}{512}$

$$\Rightarrow \frac{a(r^n - 1)}{r - 1} = \frac{3069}{512}$$

$$\Rightarrow \frac{3\left(\left(\frac{1}{2}\right)^n - 1\right)}{\left(\frac{1}{2} - 1\right)} = \frac{1023}{512}$$

$$\Rightarrow \frac{1}{2^n} - 1 = \frac{1023}{512}$$

$$\Rightarrow \frac{1 - 2^n}{2^n} = \frac{1023}{512}$$

$$\Rightarrow 2\left(1 - \frac{1}{2^n}\right) = \frac{1023}{512}$$

$$\Rightarrow 1 - \frac{1}{2^n} = \frac{1023}{1024}$$

$$\Rightarrow \frac{1}{2^n} = 1 - \frac{1023}{1024}$$

$$\frac{3069}{512}$$

$$\Rightarrow \frac{1}{2^n} = \frac{1024 - 1023}{1024}$$

$$\Rightarrow \frac{1}{2^n} = \frac{1}{1024} = \frac{1}{2^{10}}$$

$$\Rightarrow 2^n = 2^{10}$$

$$\Rightarrow n = 10$$

Hence the result.

$$a^x = a^y$$

$$\Rightarrow x = y, a \neq 0, -1, 1$$

EXAMPLE

$$\left[\frac{a}{r}, a, ar \right]$$

The sum of first three terms of a G.P. is $\frac{13}{12}$ and their product is -1 . Find the common ratio and the terms.

(Sol) Suppose that $\frac{a}{r}$, a and ar be the given first three terms of the G.P. Then;

$$\frac{a}{r} + a + ar = \frac{13}{12}$$

$$\Rightarrow a \left(\frac{1}{r} + 1 + r \right) = \frac{13}{12}$$

$$\Rightarrow a \left(\frac{r^2 + r + 1}{r} \right) = \frac{13}{12} \quad \text{--- (1)}$$

$$\text{Also } \frac{a}{r} \cdot a \cdot ar = -1$$

$$\Rightarrow a^3 = -1 \text{ or } a = -1 \quad \text{--- (2)}$$

Using (2) in (1), we get;

$$-\left(\frac{r^2 + r + 1}{r} \right) = \frac{13}{12}$$

$$\Rightarrow r^2 + r + 1 = -\frac{13r}{12}$$

$$\Rightarrow 12r^2 + 12r + 12 + 13r = 0$$

$$\Rightarrow 12r^2 + 25r + 12 = 0$$

$$\Rightarrow 12r^2 + 16r + 9r + 12 = 0$$

$$\Rightarrow 4r(3r+4) + 3(3r+4) = 0$$

$$\Rightarrow (3r+4)(4r+3) = 0$$

$$\Rightarrow r = -\frac{4}{3}, -\frac{3}{4}$$

So the required terms are;

$$-\frac{1}{4}, -1, -1 \times -\frac{4}{3}$$

$$-\frac{3}{4}, -1, \frac{4}{3} \text{ or } \frac{4}{3}, -1, -\frac{3}{4}$$

EXAMPLE

(PYQ + important)

Find the sum of the sequence $7, 77, 777, 7777, \dots$ to n terms.

(Sol) let $S_n = 7 + 77 + 777 + 7777 + \dots$ upto n Terms.

$$\Rightarrow S_n = 7(1 + 11 + 111 + 1111 + \dots \text{upto } n \text{ terms})$$

$$\Rightarrow S_n = \frac{7}{9}(9 + 99 + 999 + 9999 + \dots \text{upto } n \text{ terms})$$

$$\Rightarrow S_n = \frac{7}{9} \left\{ (10-1) + (10^2-1) + (10^3-1) + (10^4-1) + \dots + (10^n-1) \right\}$$

$$\Rightarrow S_n = \frac{7}{9} \left\{ (10 + 10^2 + 10^3 + \dots + 10^n) - n \right\}$$

$$\Rightarrow S_n = \frac{7}{9} \left\{ \frac{10(10^n-1)}{10-1} - n \right\} = \frac{7}{9} \left\{ \frac{10^{n+1} - 10 - 9n}{9} \right\}$$

$$\Rightarrow S_n = \frac{7}{81} (10^{n+1} - 9n - 10) \quad \text{Hence the result.}$$

Arithmetic Mean (A.M.) and Geometric Mean (G.M.)



★ **Arithmetic Mean:** Suppose that a and b be any positive real numbers, then their Arithmetic Mean is denoted and defined as; $A = \frac{a+b}{2}$.

★ **Geometric Mean:** Suppose that a and b be any positive real numbers, then their Geometric Mean is denoted and defined as;
 $G = \sqrt{ab}$

★ **Relationship b/w A and G:**

For any positive real numbers a & b , we know that
 $(\sqrt{a} - \sqrt{b})^2 \geq 0$ with equality if $a = b$.

$$\Rightarrow \sqrt{a}^2 + \sqrt{b}^2 - 2\sqrt{ab} \geq 0$$

$$\Rightarrow a + b \geq 2\sqrt{ab}$$

$$\Rightarrow \left(\frac{a+b}{2}\right) \geq \sqrt{ab}$$

$$\Rightarrow \boxed{A \geq G} \text{ with equality when } a = b.$$

EXAMPLE



Insert three numbers between 1 and 256 so that the resulting sequence is a G.P.

Important.

Insertion of n G.M.'s
b/w a & b .

Homework.

EXAMPLE

(Important)

If A.M. and G.M. of two positive numbers a and b are 10 and 8, respectively, find the numbers.

(Sol) According to Question;

$$A = 10 \text{ \& } G = 8.$$

$$\Rightarrow \frac{a+b}{2} = 10 \text{ \& } \sqrt{ab} = 8.$$

$$\Rightarrow a+b=20 \text{ \& } \cancel{\sqrt{ab}}^2 = 8^2$$

$$\Rightarrow \boxed{b=20-a} \text{---(1) \& } \boxed{ab=64} \text{---(2)}$$

Using (1) in (2), we get;

$$a(20-a) = 64$$
$$\Rightarrow 20a - a^2 - 64 = 0.$$

$$\Rightarrow a^2 - 20a + 64 = 0.$$

$$\Rightarrow (a-16)(a-4) = 0$$

$$\Rightarrow a = 16, 4$$

So from (1), we get

$$b = 4, 16.$$

Therefore the required numbers are

$$16 \text{ \& } 4$$

Hence the result.

EXAMPLE

If $\frac{a^{n+1} + b^{n+1}}{a^n + b^n}$ is the A.M. between a and b . Then, find the value of n .
(H/W Important)



EXAMPLE



The 5th, 8th and 11th terms of a G.P. are p , q and s , respectively. Show that $q^2 = ps$.

Proof: Given that $a_5 = p \Rightarrow ar^4 = p$ $\Rightarrow ps = (ar^4)^2 = a^2r^8$
 $a_8 = q \Rightarrow ar^7 = q$ $\Rightarrow q^2 = ps$
 $a_{11} = s \Rightarrow ar^{10} = s$ Hence proved.

Now, Consider the following:

$$ps = ar^4 \times ar^{10}$$

$$\Rightarrow ps = a^2r^{14}$$

QUESTION



Which term of the following sequences:

- (a) $2, 2\sqrt{2}, 4, \dots$ is 128? $a_n = 128$
 $a_{n-1} = 128$
- (b) $\sqrt{3}, 3, 3\sqrt{3}, \dots$ is 729?
- (c) $\frac{1}{3}, \frac{1}{9}, \frac{1}{27}, \dots$ is $\frac{1}{19683}$?

QUESTION



For what values of x , the numbers $-\frac{2}{7}, x, -\frac{7}{2}$ are in G.P.?

$$\begin{aligned} \text{H.} \quad & \frac{x}{\left(-\frac{2}{7}\right)} = \frac{\left(-\frac{7}{2}\right)}{x} \\ \Rightarrow & x^2 = \frac{-7}{2} \times \frac{-2}{7} \\ \Rightarrow & x^2 = 1 \\ \Rightarrow & x = \pm 1. \end{aligned}$$

QUESTION



Evaluate $\sum_{k=1}^{11} (2 + 3^k)$.

H/W

QUESTION

How many terms of G.P. $3, 3^2, 3^3, \dots$ Are needed to give the sum 120?

h/w

QUESTION



The sum of first three terms of a G.P. is 16 and the sum of the next three terms is 128.
Determine the first term, the common ratio and the sum to n terms of the G.P.

h/w

QUESTION



Find a G.P. for which sum of the first two terms is -4 and the fifth term is 4 times the third term.

p/w

QUESTION



If the 4th, 10th and 16th terms of a G.P. are x , y and z , respectively. Prove that x , y , z are in G.P.

||w

QUESTION

(Important) (PYQ JKBOSE)

Find the sum to n terms of the sequence, 8, 88, 888, 8888.....

QUESTION



Find four numbers forming a geometric progression in which the third term is greater than the first term by 9, and the second term is greater than the 4th by 18.

(Imp)

QUESTION

(PYQ JKBOSE)

If the p^{th} , q^{th} and r^{th} terms of a G.P. are a , b and c , respectively. Prove that

$$a^{q-r} b^{r-p} c^{p-q} = 1.$$

Proof:

Given that $a_p = a \Rightarrow AR^{p-1} = a$

$$a_q = b \Rightarrow AR^{q-1} = b$$

$$a_r = c \Rightarrow AR^{r-1} = c.$$

Now Consider the following:

$$a^{q-r} b^{r-p} c^{p-q} = (AR^{p-1})^{q-r} (AR^{q-1})^{r-p} (AR^{r-1})^{p-q}$$

$$= \frac{a^{q-r}}{A^{q-r} R^{(p-1)(q-r)}} \cdot \frac{b^{r-p}}{A^{r-p} R^{(q-1)(r-p)}} \cdot \frac{c^{p-q}}{A^{p-q} R^{(r-1)(p-q)}}$$

$$= \frac{a^{q-r} b^{r-p} c^{p-q}}{A^{q-r+r-p+p-q} R^{(p-1)(q-r) + (q-1)(r-p) + (r-1)(p-q)}}$$

$$= \frac{a^{q-r} b^{r-p} c^{p-q}}{A^0 R^{p-q+r-p+q-r+p-q-r+p-q}}$$

$$= \frac{a^{q-r} b^{r-p} c^{p-q}}{A^0 R^0} = 1 \times 1 = 1.$$

$$\Rightarrow a^{q-r} b^{r-p} c^{p-q} = 1.$$

Hence proved.

QUESTION



If the first and the n^{th} term of a G.P. are a and b , respectively, and if P is the product of n terms, prove that $P^2 = (ab)^n$.

(Imp)

QUESTION



Show that the ratio of the sum of first n terms of a G.P. to the sum of terms from $(n + 1)^{\text{th}}$

$(n + 1)^{\text{th}}$ to $(2n)^{\text{th}}$ term is $\frac{1}{r^n}$.

(Imp.)

QUESTION

(PYQ JKBOSE)

If a, b, c and d are in G.P. show that $(a^2 + b^2 + c^2)(b^2 + c^2 + d^2) = (ab + bc + cd)^2$.

Proof: Since a, b, c and d are in G.P., then

$$a = a$$

$$b = ar$$

$$c = ar^2$$

$$d = ar^3$$

Now, Consider the following;

$$(a^2 + b^2 + c^2)(b^2 + c^2 + d^2) = (a^2 + a^2r^2 + a^2r^4) \\ \times (a^2r^2 + a^2r^4 + a^2r^6)$$

$$\Rightarrow (a^2 + b^2 + c^2)(b^2 + c^2 + d^2) = a^2(1 + r^2 + r^4) \cdot a^2r^2(1 + r^2 + r^4)$$

$$\Rightarrow (a^2 + b^2 + c^2)(b^2 + c^2 + d^2) = a^4r^2(1 + r^2 + r^4)^2 \text{ --- (1)}$$

$$\text{Also } (ab + bc + cd)^2 = (a^2r + a^2r^3 + a^2r^5)^2 \\ = (a^2r(1 + r^2 + r^4))^2$$

$$\Rightarrow (ab + bc + cd)^2 = a^4r^2(1 + r^2 + r^4)^2 \text{ --- (2)}$$

From (1) & (2), we get;

$$(a^2 + b^2 + c^2)(b^2 + c^2 + d^2) = (ab + bc + cd)^2$$

Hence the result.

QUESTION

Insert two numbers between 3 and 81 so that the resulting sequence is G.P.

Imp. (H/W)

QUESTION

(PYQ JKPSC)

Find the value of n so that

$$\frac{a^{n+1} + b^{n+1}}{a^n + b^n}$$

may be the geometric mean between a and b .

(Sol) Suppose that $\frac{a^{n+1} + b^{n+1}}{a^n + b^n}$ is the G.M of a and b

$$\text{Then } \frac{a^{n+1} + b^{n+1}}{a^n + b^n} = \sqrt{ab} = (ab)^{\frac{1}{2}}$$

$$\Rightarrow \frac{a^{n+1} + b^{n+1}}{a^n + b^n} = a^{\frac{1}{2}} b^{\frac{1}{2}}$$

$$\Rightarrow a^{n+1} + b^{n+1} = a^{\frac{1}{2}} b^{\frac{1}{2}} (a^n + b^n)$$

$$a^x = a^y$$

$$x = y \quad a \neq 0, -1, 1.$$

$$\Rightarrow a^{n+1} + b^{n+1} = a^{\frac{n+1}{2}} b^{\frac{1}{2}} + a^{\frac{1}{2}} b^{\frac{n+1}{2}}$$

$$\Rightarrow a^{n+1} - a^{\frac{n+1}{2}} b^{\frac{1}{2}} = a^{\frac{1}{2}} b^{\frac{n+1}{2}} - b^{n+1}$$

$$\Rightarrow a^{\frac{n+1}{2}} (a^{\frac{1}{2}} - b^{\frac{1}{2}}) = b^{\frac{n+1}{2}} (a^{\frac{1}{2}} - b^{\frac{1}{2}}) \quad (a \neq b)$$

$$\Rightarrow a^{\frac{n+1}{2}} = b^{\frac{n+1}{2}}$$

$$\Rightarrow \left(\frac{a}{b}\right)^{\frac{n+1}{2}} = \frac{b^{\frac{n+1}{2}}}{a^{\frac{n+1}{2}}} = 1$$

$$\Rightarrow \left(\frac{a}{b}\right)^{\frac{n+1}{2}} = \left(\frac{a}{b}\right)^0$$

$$\Rightarrow \frac{n+1}{2} = 0$$

$$\Rightarrow \boxed{n = -1}$$

Hence the result.

QUESTION



The sum of two numbers is 6 times their geometric mean, show that numbers are in the ratio $(3 + 2\sqrt{2}) : (3 - 2\sqrt{2})$.

(Imp.)

QUESTION



If A and G be A.M. and G.M., respectively between two positive numbers, prove that the

numbers are $A \pm \sqrt{(A + G)(A - G)}$

(H/W) (Imps).

QUESTION



$\forall$ (for all)

If f is a function satisfying $f(x+y) = f(x)f(y)$ for all $x, y \in \mathbb{N}$ such that $f(1) = 3$ and

$$\sum_{x=1}^n f(x) = 120, \text{ find the value of } n.$$

(Sol) Given that $f(x+y) = f(x) * f(y) \quad \forall x, y \in \mathbb{N}$.

$$f(1) = 3$$

$$\sum_{x=1}^n f(x) = 120.$$

$$\Rightarrow f(1) + f(2) + f(3) + \dots + f(n) = 120 \quad \text{--- (1)}$$

$$\text{Here } f(2) = f(1+1) = f(1) \cdot f(1) = 3^2.$$

$$f(3) = f(2+1) = f(2) \cdot f(1) = 3^3$$

$$f(4) = f(3+1) = f(3) \cdot f(1) = 3^4$$

$$\vdots$$
$$f(n) = 3^n.$$

Therefore from (1), we get;

$$3 + 3^2 + 3^3 + 3^4 + \dots + 3^n = 120.$$

$$\Rightarrow \frac{3(3^n - 1)}{3 - 1} = 120.$$

$$\Rightarrow 3(3^n - 1) = 120 \times 2.$$

$$\Rightarrow 3^n - 1 = \frac{120 \times 2}{3}$$

$$\Rightarrow 3^n - 1 = 80$$

$$\Rightarrow 3^n = 81 = 3^4.$$

$$\Rightarrow n = 4.$$

Hence the result

QUESTION



If $\frac{a + bx}{a - bx} = \frac{b + cx}{b - cx} = \frac{c + dx}{c - dx}$ ($x \neq 0$), then show that a, b, c and d are in G. P.

Imp.
(PYQ JKBOSE)

QUESTION



If a, b, c, d are in G.P, prove that $(a^n + b^n), (b^n + c^n), (c^n + d^n)$ are in G.P.

(PYQ JKBOSE
Imp.)

Thank
You